Module 1

1. Explain classification of cybercrime with the help of examples.

2. Analyze the various factors that contribute to the creation of vulnerabilities in cyber systems.

3. Explain Types of Cybercriminals.

4. Analyse the concepts of vulnerabilities, threats, and risks in the context of information security.

5. Evaluate the importance of the CIA Triad in cybersecurity.

6. Write a short note on Internet Governance.

7. Difference between Passive and Active attack.

8. Describe the Need for a Comprehensive Cyber Security Policy.

Module 2

1. Explain DES algorithm with block diagram.

2. Explain RSA algorithm with its application.

3. Explain the concept of cryptography and illustrate its need by applying it to a real-life scenario with a suitable example.

4. Compare the AES and DES cryptography algorithms in terms of their structure, key sizesand security features.

5. Explain Asymmetric Key Cryptography with suitable example.

6. Explain Diffie hellman key exchange algorithm with its application.

7. Difference between Symmetric and Asymmetric Key Cryptography.

8. Types of cipher.

9. Difference between block cipher and stream cipher.

Module 3

Explain Cyber Security Safeguards and its types in detail.
Describe Cyber Security Audit Process.
Differentiate between different types of firewalls.
Types of Access Control.
Access Control models and mechanisms.
Describe Cyber Security Audit types.
What is mean by Authentication? Describe the different authentication models.
What are the different threats in authentication? What different practices one should follow for authentication?
Different types of Biometrics.
Different advantages and limitations in Biometrics.
Applications of biometrics.
Describe Denial of Service Attack in detail.
Explain in detail about ethical hacking.
List down the advantages, limitations and applications of ethical hacking.
Explain Intrusion Detection System (IDS). Differentiate between types of IDS.
Explain threat management process.
Explain relationshipSecurity Policy & Threat Management.
Explain Security policy and its objectives.
Types of Security Policies and its key components.

Module 4

Explain Cybercrime Scene Investigation Process with the help of diagram.
Describe process of Identifying and Documenting the Scene & Evidence of Cyber Crime Scene with an example.
Demonstrate Chain of custody form with an example.
Digital forensic lifecycle with neat labelled diagram.
Challenges and applications of digital forensics.
Chain of custody process
Write a short note on Forensic Imaging.
Describe Steps in Cloning.
Difference between Live System Forensics and Dead System ForensicsLive System Forensics
Different Techniques and Methods in Preservation of evidence.
Write a short note on hashing.
Describe Process of Using Hashing in Evidence Handling.
Explain Examination of Evidence and task involved in it.
Explain Analysis of Evidence and task involved in it.
Write a short note on Report Generation in Digital Forensics with sample of Report Structure.
Key components in Report Generation in Digital Forensics.

Module 5

Explain the role of metadata analysis in digital forensics.

Describe how Windows registry examination helps during a forensic investigation.

Differentiate between disk imaging and data recovery with suitable examples.

Explain the purpose of Restore Points and Shadow Copies in retrieving system information.

What is the significance of analyzing the Recycle Bin during digital forensic analysis?

Illustrate the function of EnCase and FTK tools in forensic investigations.

Explain anti-forensics techniques and their impact on digital investigations.

Describe vulnerability assessment tools and how they support digital forensics.

Explain the steps involved in examining tampered/deleted data.

Discuss the role of data recovery tools in retrieving user information from storage devices.

Analyze how metadata can be used to trace file activity in a cybercrime case.

Demonstrate how a forensic investigator can retrieve deleted files using Restore Points and Shadow Copies.

Apply the procedure of Windows registry analysis to track USB usage history.

Compare the features and capabilities of EnCase and FTK tools in forensic investigations.

Analyze a scenario where the Recycle Bin is emptied intentionally—identify possible forensic recovery methods.

Explain how disk imaging can support the chain of custody in digital investigations.

Evaluate different anti-forensic methods and propose counter-techniques to overcome them.

Given a tampered hard disk, outline the process of recovering maximum possible data using forensic tools.

Examine the role of vulnerability assessment tools in identifying cyber-attack sources during forensic analysis.

Investigate a case where partial data wiping is performed and recommend the best data recovery tools for evidence retrieval.

Module 6

Explain the importance of Cyber Security Regulations in protecting digital infrastructure.

Describe the role of International Law in dealing with cross-border cybercrimes.

Explain the responsibilities of the State in maintaining security within cyberspace.

Discuss the role of the Private Sector in safeguarding cyberspace.

Describe the need for Cyber Security Standards in organizations.

Explain the characteristics of the Indian Cyberspace environment.

Summarize the key objectives of the National Cyber Security Policy (NCSP) 2013.

Explain the significance of the Indian Evidence Act in investigating cybercrime cases.

Describe the relevance of the IPC and Criminal Procedure Code (CrPC) to cyber offences.

Explain the main purpose of the Electronic Communication Privacy Act (ECPA).

Discuss the term "Legal Policies" in the context of cyber security.

Differentiate between national and international cyber security regulations with examples.

Analyze how international cyber laws help resolve cybercrimes involving multiple countries.

Examine a cyber-attack scenario and determine the role of the State versus the Private Sector in mitigating the impact.

Apply cyber security standards (such as ISO/IEC 27001) to ensure secure data handling in an organization.

Evaluate the effectiveness of the National Cyber Security Policy 2013 in strengthening cybersecurity in India.

Analyze the role of the Indian Evidence Act in presenting digital evidence in court.

Compare how the IPC and CrPC deal with cyber offences and suggest improvements.

Investigate a case of email privacy violation under the Electronic Communication Privacy Act (ECPA) and suggest applicable legal actions.

Examine the challenges faced by India in regulating cyberspace and propose solutions.

Apply cyber security regulatory frameworks to safeguard critical information infrastructure in India.

Assess how legal policies influence organizational adoption of cybersecurity practices.